

Subject Description Form

Subject Code	LSGI4701
Subject Title	Smart Cities Project Management
Credit Value	2
Level	4
Pre-requisites	LSGI2801 Smart Cities: Context, Policy and Government
Objectives	This subject aims to nurture the students' ability to manage the smart city projects which utilize urban informatics technologies (urban sensing, urban computing, and urban big data infrastructure). The students will learn and practice the essentials of the smart city project management, including the key procedures in a smart city project, technical perspective on adopting the urban informatics technologies and using data infrastructure for project management, documentation of the project, and communicating with other parties in the project life cycle.
Intended Learning Outcomes	After learning this subject, students are expected to be able to: 1. Understand the contemporary environment and opportunities for smart city business in Hong Kong (L2) 2. Perform the key procedures of conducting a smart city project (L3) 3. Produce proper documentation of a smart city project (L4) 4. Communicate and compromise with other professionals (e.g., government department officers, developers of urban informatics technologies used in the project) in a smart city project life cycle (L4) 5. Apply principles of project accounting to manage budgeting, costing, and financial control in smart city projects. (L3)
Subject Synopsis/ Indicative Syllabus	A. Smart City Projects / Business • Types of smart city project based on the primary technologies involved: urban geomatics, Internet of Things, artificial intelligence, remote sensing, etc. • Operations management of smart city projects B. Management of Smart City Projects • Feasibility study • Theme or objective identification, • Tendering process • Project development lifecycle

	• Project management methodology C. Urban spatial big data infrastructure as a project management tool • Using urban spatial big data infrastructure (e.g., Building Information Model) for project management, and environmental impact analysis • Achieving interoperability among service providers in the project • Handling legal and policy issues D. Project management accounting: principles and procedures • Accounting fundamentals for PMs • Cost estimation and budgeting in smart city project • Pricing strategies and tender quotation • Financial control and monitoring during project execution						
Teaching/Learning Methodology	The course is designed to adopt interactive lectures such that the basic information of each lecture is made available on-line. The interactive lecture starts by introducing essential basics and users group activity and debriefing to encourage active thinking and participation. Hybrid PBL and Case-based learning are used increasingly in the lecture to work on local and international smart city cases.						
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
			1	2	3	4	5
	1. Quiz	15	✓	✓	✓	✓	
	2. Essay	30	✓	✓	✓		✓
	3. Smart city group project	55		✓	✓	✓	✓
	Total	100					

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

- Quiz(zes) to test students' basic knowledge on project management taught in the class.
- An individual essay which asks students to conduct literature review and analysis of smart city projects, to let the students aware the current development and chances of smart city projects in and beyond Hong Kong.
- A group project working on a specific topic of smart city development to let students exercise the essential procedures and documentation of the project.

Student Study Effort Expected	Class contact:	
	▪ Lecture/tutorial	26 Hrs.
	Other student study effort:	
	▪ literature review and web surfing	10 Hrs.
	▪ group project	30 Hrs.
	▪ self-study and revision	10 Hrs.
	Total student study effort	76 Hrs.
Reading List and References	1. Shi, W., Goodchild, M., Batty, M., Kwan, M.-P., Zhang, A. (Eds.) (2021), <i>Urban Informatics</i>, Springer. 2. Russell, S. J. (2010). <i>Artificial intelligence a modern approach</i>, Pearson Education, Inc.. 3. Greengard, S. (2021). <i>The internet of things</i>. MIT press. 4. Zimmerer, T. and Scarborough, N. (2005), <i>Essentials of Entrepreneurship and Small Business Management</i>, 4th ed., Pearson Education. 5. Course, L., Hubbard, D. and Wong, E. (2003), <i>Butterworths Hong Kong Contract Law Handbook</i>, Hong Kong: Lexis Nexis Butterworths. 6. Willoughby, P. and Wilkinson, M. (1995), <i>Registration of Titles in Hong Kong</i>, Hong Kong: Butterworth. 7. Harmon, J. (2003), <i>The Design and Implementation of Geographic Information Systems</i>, John Wiley & Sons. 8. Huxhold, W. (1994), <i>Managing Geographic Information System Projects</i>, New York: Oxford University Press. 9. HKEDC (1996, http://www.icac.org.hk/hkedc), Ethics for Professionals (Architecture, Engineering & Surveying), Hong Kong Ethics Development Centre, ICAC, HKSAR. 10. OCGIO (2012) <i>User Guide on PRINCE</i>, Office of the Government Chief Information Officer, HKSAR 11. OCGIO (2012) <i>An Introduction to Structured Systems Analysis & Design Methodology (SSADM)</i>, Office of the Government Chief Information Officer, HKSAR 12. Ruben de Laat and Leon Van Berlo (2011) <i>Integration of BIM and GIS: The Development of CityGML GeoBIM Extension</i>, <i>Advances in 3D Geo-Information Science</i>, Berlin, Springer. pp211-226.	

	13. CIC (2015) <i>Hong Kong BIM Standards Building Information Model - Project Execution Plan (BIM PEP) Construction Industry Council</i>, HKSAR. 14. Accountants, C. P., & Bureau, B. B. (2011). <i>Project Management Accounting: Budgeting, Tracking, and Reporting Costs and Profitability</i>, by Kevin R. Callahan, Gary S. Stetz and Lynne M. Brooks Copyright© 2011 John Wiley & Sons, Inc. balance, 64(65), 143. 15. Kerzner, H. (2025). <i>Project management: a systems approach to planning, scheduling, and controlling</i>. John Wiley & Sons. 16. Meredith, J. R., Shafer, S. M., & Mantel Jr, S. J. (2017). <i>Project management: a strategic managerial approach</i>. John Wiley & Sons. 17. Ogaja, C. (2016). <i>Geomatics engineering: a practical guide to project design</i>. CRC Press.
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